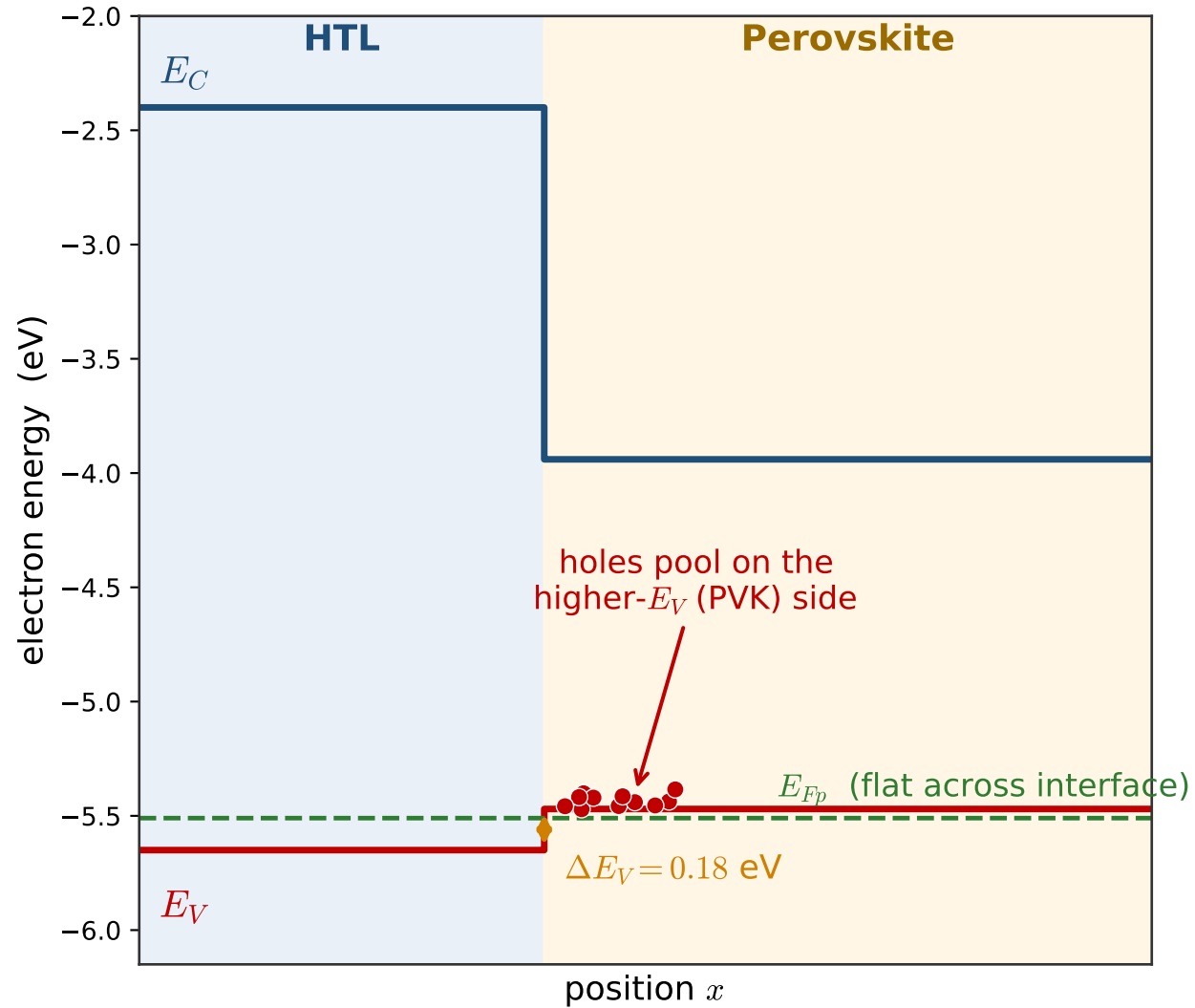


Why holes pile up at the HTL / Perovskite heterointerface



1. Valence-band edge of each layer

$$E_V = -(\chi + E_g)$$

$$\text{HTL: } -(2.40 + 3.25) = -5.65 \text{ eV}$$

$$\text{PVK: } -(3.94 + 1.53) = -5.47 \text{ eV}$$

2. Valence-band offset (the ledge)

$$\Delta E_V = E_V^{\text{PVK}} - E_V^{\text{HTL}} = 0.18 \text{ eV}$$

3. Boltzmann hole density

$$p = N_V \exp\left(\frac{E_V - E_{Fp}}{kT}\right)$$

4. Pile-up ratio across the step (E_{Fp} cancels)

$$\frac{p_i}{p_{i-1}} = \exp\left(\frac{\Delta E_V}{kT}\right) = \exp\left(\frac{0.18}{0.0259}\right) \approx 1000$$